Mahalanobis Distance

The Mahalanobis distance of a point $\boldsymbol{x} = (x_1, x_2, \ldots, x_n)^\top \in \mathbb{R}^n$ from a probability distribution:

$$d = \sqrt{(\boldsymbol{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\boldsymbol{x} - \boldsymbol{\mu})}$$

with mean vector $\boldsymbol{\mu} \in \mathbb{R}^n$ and covariance matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{n \times n}$.

Wikipedia page	Page view (Desktop)	Page view (Mobile)
Machine_learning	88,260	56,966
Deep_learning	36,741	15,321
Statistics	29,673	28,341
Mathematical_optimization	19,318	6,819
Data_mining	13,168	5,750
Supervised_learning	14,120	3,596
Unsupervised_learning	11,693	2,702
Computer_program	13,451	16,587
Algorithm	38,905	36,203
Reinforcement_learning	30,393	9,841
Pattern_recognition	6,119	3,223
Computer_science	47,723	40,321
Probability_theory	10,913	8,788
Regression_analysis	32,814	16,207
Statistical_classification	3,947	1,403
Anomaly_detection	7,540	1,514

The dataset is available at https://arxiv.org/abs/2605.16361

$$\text{Mean vector } \boldsymbol{\mu} = \begin{bmatrix} 2.5 \times 10^4 \\ 1.6 \times 10^4 \end{bmatrix}$$

$$\text{Covariance matrix } \boldsymbol{\Sigma} = \begin{bmatrix} 4.6 \times 10^8 & 3.2 \times 10^8 \\ 3.2 \times 10^8 & 2.7 \times 10^8 \end{bmatrix}$$

Covariance Matrix $\mathbf{\Sigma} \in \mathbb{R}^{2\times 2}$

Eigenvalue decomposition:

$$\mathbf{\Sigma} = \lambda_1 \boldsymbol{u}_1 \boldsymbol{u}_1^\top + \lambda_2 \boldsymbol{u}_2 \boldsymbol{u}_2^\top$$

with eigenvalues:

$$\lambda_1 = 6.98 \times 10^8$$
$$\lambda_2 = 0.28 \times 10^8$$

and eigenvectors:

$$\boldsymbol{u}_1 = \begin{bmatrix} 0.8 \\ 0.6 \end{bmatrix} \quad \boldsymbol{u}_2 = \begin{bmatrix} -0.6 \\ 0.8 \end{bmatrix}$$

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```python
import numpy as np

S = np.array([[4.57602874e+08, 3.21703384e+08], [3.21703384e+08, 2.68455504e+08]])
eig_val, eig_vec = np.linalg.eig(S)

print('lambda_1 = ', eig_val[0])
print('lambda_2 = ', eig_val[1])
print('u_1 = ', eig_vec[:, 0])
print('u_2 = ', eig_vec[:, 1])
```

The squared Mahalanobis distance is an ellipse in the two-dimensional space.

$$
\begin{aligned}
d^2 &= (\boldsymbol{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\boldsymbol{x} - \boldsymbol{\mu}) \\
&= (\boldsymbol{x} - \boldsymbol{\mu})^\top \left(\lambda_1 \boldsymbol{u}_1 \boldsymbol{u}_1^\top + \lambda_2 \boldsymbol{u}_2 \boldsymbol{u}_2^\top \right)^{-1} (\boldsymbol{x} - \boldsymbol{\mu})
\end{aligned}
$$

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$$
\begin{aligned}
d^2 =& (\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\boldsymbol{x}-\boldsymbol{\mu}) \\
=& (\boldsymbol{x}-\boldsymbol{\mu})^\top \left(\lambda_1 \boldsymbol{u}_1 \boldsymbol{u}_1^\top + \lambda_2 \boldsymbol{u}_2 \boldsymbol{u}_2^\top\right)^{-1} (\boldsymbol{x}-\boldsymbol{\mu}) \\
=& (\boldsymbol{x}-\boldsymbol{\mu})^\top \left(\frac{1}{\lambda_1} \boldsymbol{u}_1 \boldsymbol{u}_1^\top + \frac{1}{\lambda_2} \boldsymbol{u}_2 \boldsymbol{u}_2^\top\right) (\boldsymbol{x}-\boldsymbol{\mu}) = c
\end{aligned}
$$

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$$
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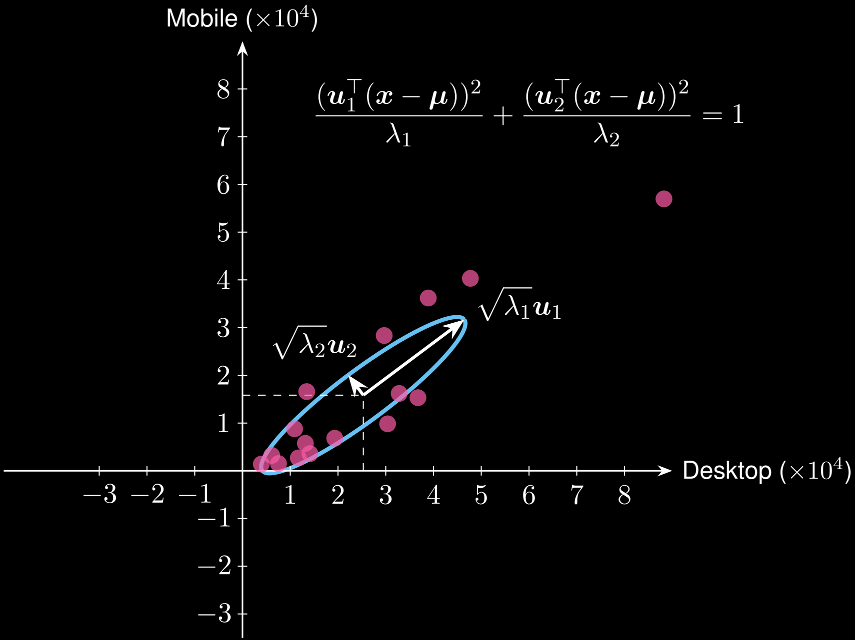
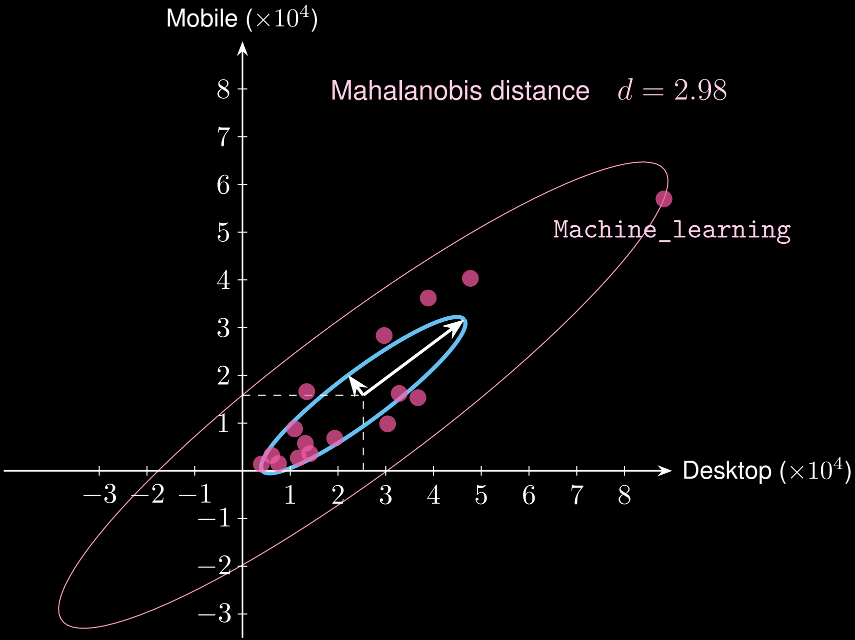
&= (\boldsymbol{x}-\boldsymbol{\mu})^\top \left(\lambda_1 \boldsymbol{u}_1 \boldsymbol{u}_1^\top + \lambda_2 \boldsymbol{u}_2 \boldsymbol{u}_2^\top\right)^{-1} (\boldsymbol{x}-\boldsymbol{\mu}) \\
&= (\boldsymbol{x}-\boldsymbol{\mu})^\top \left(\frac{1}{\lambda_1} \boldsymbol{u}_1 \boldsymbol{u}_1^\top + \frac{1}{\lambda_2} \boldsymbol{u}_2 \boldsymbol{u}_2^\top\right) (\boldsymbol{x}-\boldsymbol{\mu}) = c
\end{aligned}
$$

$$
\Rightarrow \qquad \frac{(\boldsymbol{u}_1^\top (\boldsymbol{x}-\boldsymbol{\mu}))^2}{\lambda_1} + \frac{(\boldsymbol{u}_2^\top (\boldsymbol{x}-\boldsymbol{\mu}))^2}{\lambda_2} = d^2 \qquad \text{(ellipse)}
$$

with eigenvalues λ_1, λ_2 and eigenvectors $\boldsymbol{u}_1, \boldsymbol{u}_2$.


Mobile $(\times 10^4)$
Desktop $(\times 10^4)$
$\frac{(\boldsymbol{u}_1^\top(\boldsymbol{x}-\boldsymbol{\mu}))^2}{\lambda_1} + \frac{(\boldsymbol{u}_2^\top(\boldsymbol{x}-\boldsymbol{\mu}))^2}{\lambda_2} = 1$
$\sqrt{\lambda_1}\boldsymbol{u}_1$
$\sqrt{\lambda_2}\boldsymbol{u}_2$


Mobile $(\times 10^4)$
Desktop $(\times 10^4)$
Mahalanobis distance $d = 2.98$
Machine_learning
8
7
6
5
4
3
2
1
−1
−2
−3
−3
−2
−1
1
2
3
4
5
6
7
8

Thanks for your attention!

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